Total No. of Questions: 05

Total No. of Printed Pages: 02

Roll No.________________

BTCS502

IV Semester Examination, May – June 2022

B.C.A(BDAI)/B.C.A.+M.C.A.(NB,BKNT)/B.Tech./B.Tech.+M.B.A./B.Tech.+M.Tech. (CSE,CCE,CSE-BDA,CSE-CF,CSE-CC,CSE-CYFS,BDCE-Impetus,CSE-ICS,CSE-MA,CSE-BDAI,CSE-CMCI,CSE-ES,CSE-DSI,CSE-FSDI,CSE-All)

OPERATING SYSTEMS

Choice Based Credit System (CBCS)

Time: 3 Hrs.

Maximum Marks: 60

Minimum Pass Marks: 24

Note: *All questions carry equal marks, out of which part'A' and 'B' carry 3 marks and part 'C' carries 6 marks.*
From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.
Draw neat diagram, wherever necessary.
Assume suitable data wherever necessary.

Q.1(A) Differentiate between Multitasking and multiprogramming OS. **03**

(B) Explain the concept of kernel & user level threads. **03**

(C) What are the various services provided by the operating system? How each service provides convenience to the users? Explain in which case it would be impossible for user level programs to provide these services. **06**

OR

What is meant by a system calls? How can it be used? How does an application program use these system calls during execution? Explain the types of system calls.

Q.2(A) What is process? Explain PCB with diagram. **03**

(B) Describe the differences among short term, medium term and long term scheduling. **03**

(C) Compare the following CPU scheduling algorithms with examples:

1. FCFS 2. SRTF 3. Round Robin **06**

OR

Consider the set of processes with arrival time (in milliseconds), CPU burst time (in milliseconds), and priority (0 is the highest priority) shown below.

Process	Arrival time	Burst Time	Priority
P1	0	11	2
P2	5	28	0
P3	12	2	3
P4	2	10	1
P5	9	16	4

Calculate the average waiting time (in milliseconds) of all the processes using preemptive priority scheduling algorithm.

Q.3(A) What do you mean by deadlock prevention? A computer has six tape drives with processes competing for them. Each process needs two tape drives for which values of n the system is deadlock free. **03**

Contd...

(B) Explain the following terms with examples:

1. Critical Section 2. Race Condition **03**

(C) What are the various operations that can be performed on a semaphore? What is the difference between binary and counting semaphores? Write the code for semaphore operations. **06**

OR

A system with following processes and resource exists:

Process	Allocation	Max	Available
	A B C	A B C	A B C
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

Answer the following questions using Banker's algorithm:

1. What will be the content of the need matrix?
2. Is the system in a safe state? If yes, what is the safe sequence?
3. If a request from process p1 arrives for (1, 0, 2) can the request be granted immediately?

Q.4(A) What are the major problems to implement demand paging? Explain. **03**

(B) What is compaction? How is it implemented? Explain. **03**

(C) What is Segmentation? Explain virtual to physical address mapping in a segmented system with the help of a diagram. **06**

OR

Consider the following page reference string

1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6

How many page faults occur for the following replacement algorithm, assuming 3 frames?

1. FIFO replacement 2. Optimal replacement

Q.5(A) Explain the concept of Master Boot Record. **03**

(B) Describe how a file directory system can be organized into one-level, two-level and tree-structure directories. **03**

(C) Discuss various file allocation and access methods. Compare their advantages and disadvantages. **06**

OR

The disk queue in the request for I/O to block on cylinders are 98,183,37,122,14,124,65,67. If the disk head is initially at 53, compute the total head movements for the following algorithms:

1. SSTF 2. SCAN

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